

# MANAS AGRAWAL

Boston, MA | (857) 265-1533 | [aggrawal.m@northeastern.edu](mailto:aggrawal.m@northeastern.edu) | [Portfolio](#) | [Linkedin](#) | [Github](#) | [Medium](#) | [Leetcode](#)

## EDUCATION

### Northeastern University

Masters of Science – Computer Science | GPA: 3.83

Boston, MA

Sep 2024 – Dec 2026

- Relevant Coursework: Algorithms, System Design, Principles of Programming Languages
- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)
- Researching Multi-Agent LLM architectures, for e2e SDLC automation, across accuracy, cost efficiency

## WORK EXPERIENCE

### Hydrow

Software Engineer Intern

Boston, MA

Jan 2026 – Present

- Built REST APIs using NestJS, PostgreSQL, and Redis, serving real-time athlete performance data across Hydrow's platform
- Engineered async task queue with bounded concurrency to safely batch-process movement records in parallel, preventing resource exhaustion across shared DB and Redis instances in a multi-container Kubernetes environment
- Optimized PostgreSQL query performance using materialized views, lateral cross joins and indexed lookups, reducing table scans and query execution time across data APIs

### canAssist × Northeastern University

Forward Deployed Engineer & Scrum Master, Research Capstone

Boston, MA

May 2026 – Present

- Bootstrapped NestJS REST backend using own open-source boilerplate, designed PostgreSQL schema, optimized performance via Redis caching, and instrumented end-to-end distributed tracing using own published telemetry npm package
- Led client engagement across requirements gathering, sprint demos, and feedback sessions while serving as Scrum Master for a 5-person team, driving sprint ceremonies, GitHub issue tracking, and daily standups

### Studio Graphene

Forward Deployed Engineer

Gurgaon, India

Nov 2020 – Jul 2024

- Built serverless engineering analytics platform using AWS Lambda to pinpoint bottlenecks like high PR wait times, frequent build failures, and blocked dependencies, reducing cycle time and boosting engineering velocity by 23%
- Built distributed, event-driven microservices using AWS SQS messaging queues for decoupled data ingestion and processing, reducing response latency by ~40%
- Ingested raw data from 10k+ weekly events from third party sources into Elasticsearch and ran aggregation queries to calculate high level metrics
- Built scalable Node.js/TypeScript REST API backend processing 2M+ API requests daily, designed PostgreSQL schema with partitioning and indexing strategies to handle 500GB+ product catalog data and real-time cross-region inventory sync
- Deployed docker containerized microservices on AWS ECS, built CodePipeline CI/CD workflows, configured CloudWatch alarms, and integrated SES for email notifications on deployment failures and system alerts
- Developed real-time tracking and competitive scoring algorithm using Node.js/PostgreSQL with read replicas and connection pooling to serve 100K+ active users, for oddchecker's betting platform

## OPEN SOURCE PROJECTS

[NestJs Backend Boilerplate](#) | *Nodejs, TypeScript, NestJs, Docker, Git, Zod, Biome, Winston*

- A production-ready boilerplate with all the essentials like JWT authentication, RBAC, validation, logger, database, Prisma ORM, Docker and Swagger OpenAPI Specs pre-configured. It has 51 stars and 11 forks so far

[Telemetry NPM Package](#) | *Nodejs, TypeScript, AWS X-Ray, npm, OpenTelemetry, cloudwatch, Jaeger, Prometheus*

- Built and published a package on npm, enabling end-to-end distributed tracing across APIs, DB queries, and async tasks thereby reducing mean debugging time by ~60%
- It has over 1000 downloads on npm

## TECHNICAL SKILLS

**Languages:**

JavaScript, TypeScript, Java, C++, Python

**Frameworks:**

Node.js, Express.js, NestJS, Django, Laravel, React

**Database:**

SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase

**Cloud & DevOps:**

Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure

**Others:**

Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, frontend, agile, SDLC